

Representation Convergence -- G8, G9, G10

Training-free similarity (G8), fractal-trained vision models (G9), and hub entry-map recoverability (G10). All results from real-cache runs, 9,533 COCO items. Post-defense ongoing research. All reviewer-identified code bugs fixed.

Reproduction gates

gate	published	measured	status
G9: convnext~img_small	0.397	0.3971	ok
G9: convnext~img_base	0.392	0.3915	ok
G9: convnext~img_large	0.391	0.3914	ok
G9: bge~sbert	0.768	0.7682	ok
G9: cross-modal mean	0.290	0.290	ok
G10: img_base % native	95.9%	94.8%	ok
G10: img_large % native	92.9%	93.4%	ok
G10: random-map control	chance	0.001	ok

Both gates **VERIFIED**.

Bug fixes applied (from ChatGPT review + own discovery)

#1 P3 bootstrap: replaced invalid bootstrap-of-30-overlapping-subsample-means with item-level paired bootstrap (m=600, B=120). P3 now reads INCONCLUSIVE with real-width CIs.

#2 drop1: fixed from Xho - (centered projection) to Xc - (Xc @ v1) @ v1. Corrected drop1 shows frac_cnn signal SURVIVES PC1 removal (was wrongly reported as destroyed).

#4 GPT-2 flag: G8 anisotropy threshold raised from 0.30 to 0.95 so only genuine collapse (GPT-2 at 0.999) flags; bge/bert/sbert restored as healthy.

#5 CKA asymmetry: whitened view now L2-row-normalized to match raw view.

G10 H3 floor guard: '% of native' is NaN when native R@1 is at the floor. H3 reads NOT MEANINGFUL instead of the false '100% of native, CONFIRMED'.

G10 find() bug: replaced Path.rglob with os.walk + fnmatch for Google Drive FUSE compatibility with '+' in filenames.

G8 -- Training-free similarity across eight encoders

Four metrics (mutual k-NN, debiased CKA, relative representations, shape-rho) across all encoder pairs, raw and whitened, with shuffle controls. Roster: DINOv2 s/b/l, ConvNeXt, bge, GPT-2, BERT, SBERT. 8/8 loaded, 0 alignment failures, only GPT-2 flagged degenerate (mean pair-cos 0.999, threshold 0.95).

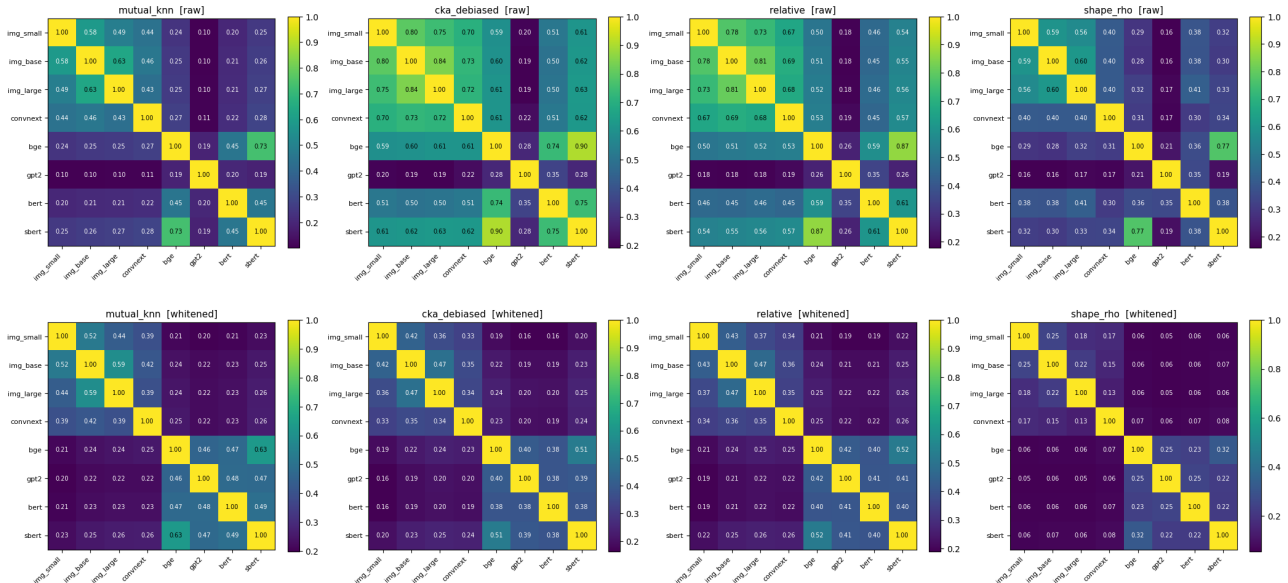


Figure G8.1 -- Four metrics x two geometries. Whitened CKA values shifted slightly from earlier runs due to the L2-row-normalization fix (#5). Cross-modal shape-rho: raw 0.288, whitened 0.062.

Key findings

Metrics agree on structure, disagree on GPT-2. gpt2~bert: shape-rho 0.346 but mutual-kNN only 0.202. The two metrics diverge where anisotropy is worst.

Whitening collapses cross-modal agreement. Stratified shape-rho: within-family 0.631 -> 0.241 (halves), cross-modal 0.288 -> 0.062 (near-zero). The C.11/C.12 reading-vs-finding tension in a third form.

Sample sizes. G8 uses 4,000-row views and 2,000 for shape-rho (subsample estimates); G9's 0.290 cross-modal mean is the full-n statistic.

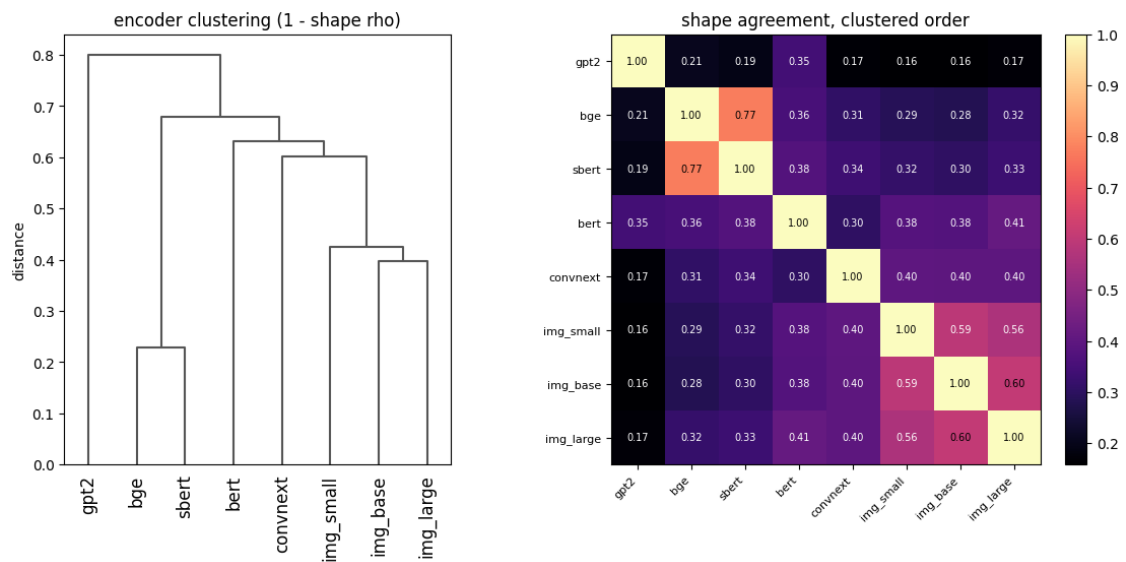


Figure G8.2 -- Hierarchical clustering on (1 - shape-rho). GPT-2 isolates at 0.80; bge-sbert tightest pair at 0.77; DINOv2 trio clusters with ConvNeXt adjacent.

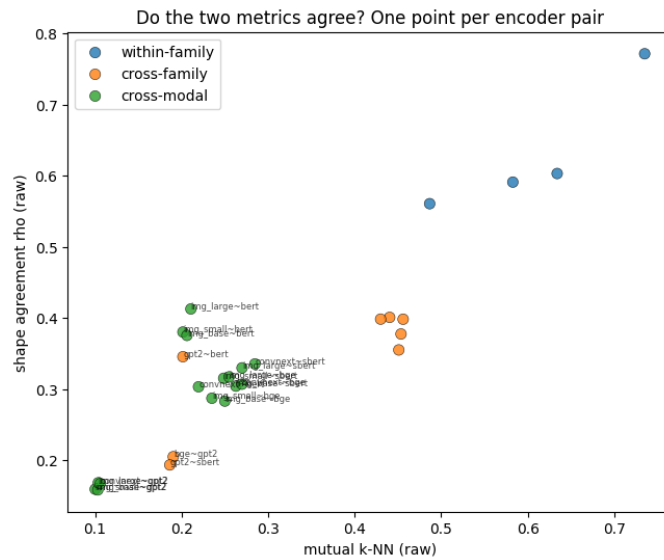


Figure G8.3 -- Metric-agreement scatter. GPT-2 pairs anchor the bottom-left.

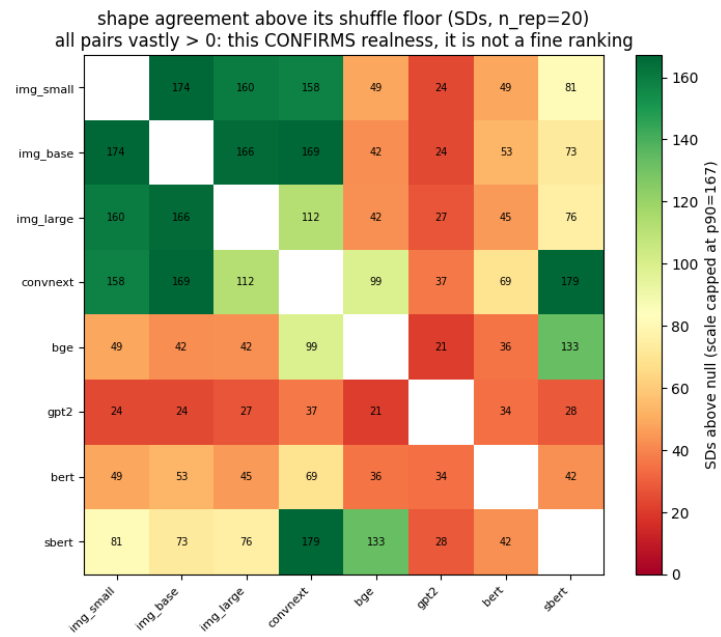


Figure G8.4 -- SDs above shuffle floor (n_rep=20, fresh samples). All pairs clear by 17-210 SDs. Confirms realness; does not rank pairs.

G9 -- Fractal-trained vision models vs the eight encoders

Two FractalDB-1k models (ResNet-50, DeiT-tiny) measured by shape-rho against all eight encoders, bracketed by random-init and ImageNet-1k twins plus a pixel floor. Reality fraction R places each on a 0-to-1 scale.

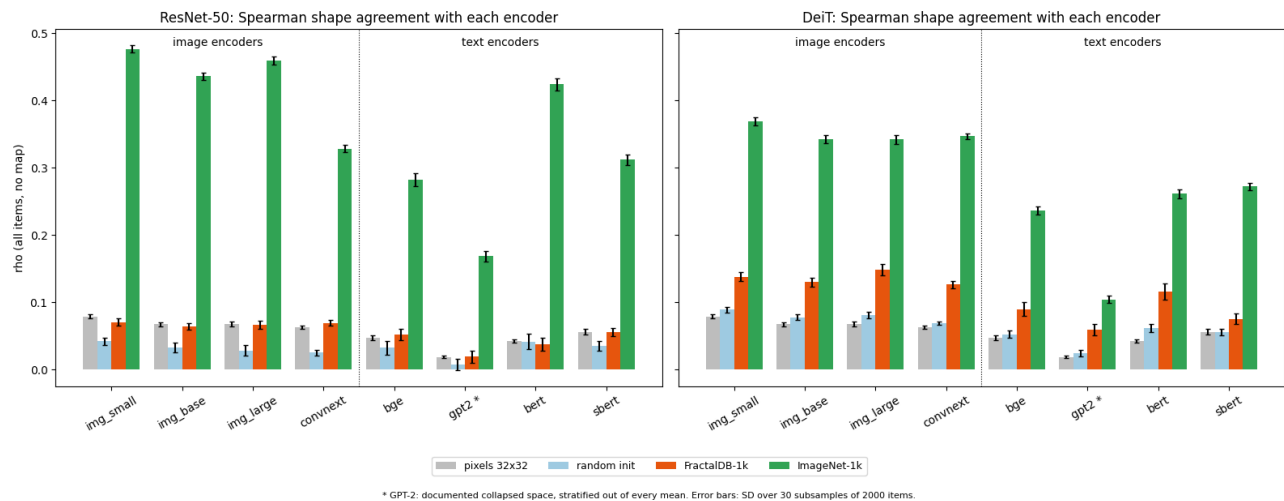


Figure G9.1 -- Shape-rho brackets. ImageNet (green) dominates; fractal (orange) sits well below but above random/pixels on the ViT. CNN fractal at the pixel floor (anisotropy).

Pre-registered verdicts (corrected)

prediction	CNN	ViT	note
P1 fractal < ImageNet	CONFIRMED 8/8	CONFIRMED 8/8	
P2 fractal > random	CONFIRMED 4/4	CONFIRMED 4/4	norm confound flagged
P3 R_image > R_text	INCONCLUSIVE	INCONCLUSIVE	item-level paired bootstrap
P5 fractal > pixels	FALSIFIED 0/4	CONFIRMED 4/4	CNN: see F.6

P3 CIs: CNN [-0.106, +0.100], ViT [-0.053, +0.135]. Wide, straddling zero -- no reliable image-vs-text asymmetry in R. The earlier tight [+0.036, +0.055] CONFIRMED was an artifact of bootstrapping overlapping subsample means (bug #1, now fixed). P3w: INCONCLUSIVE (CNN [-0.066, +0.283], ViT [-0.165, +0.121]).

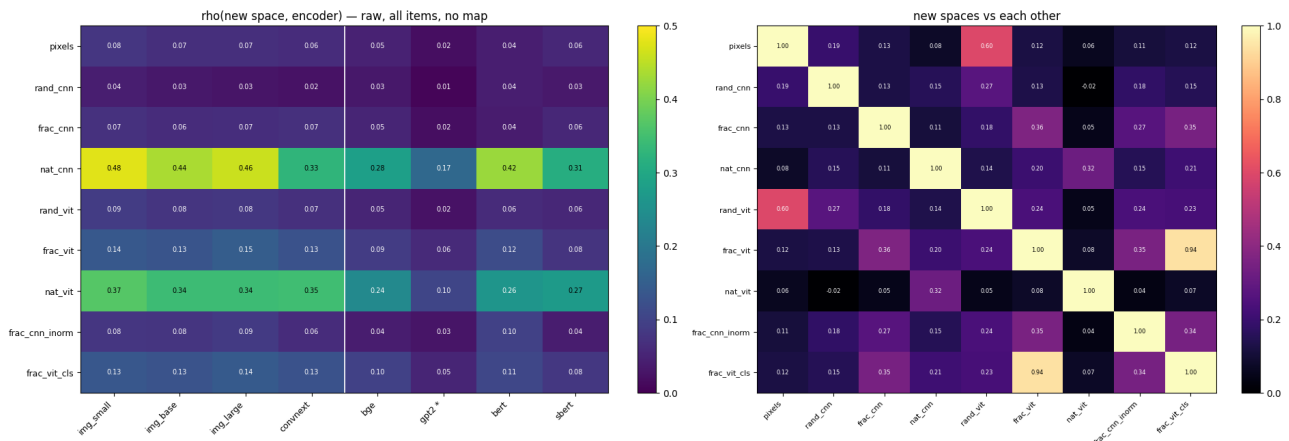


Figure G9.2 -- Left: raw shape-rho (bracket spaces x 8 encoders). Natural rows bright, fractal dim, GPT-2 column marked. Right: bracket spaces against each other.

F.6 -- Whitening rescue + corrected drop1

P5w-cnn flips to CONFIRMED 4/4 after whitening. $R_{white} \sim 11\%$ of the natural-image bracket. Small but real.

Corrected drop1: frac_cnn at 0.060 vs pixels 0.051, random 0.051 -- ordering **holds** (previously falsely reported as destroyed due to the centering bug). The CNN fractal signal survives PC1 removal.

ViT drop1 differs: frac_vit drops to 0.067 vs random 0.098 -- ViT signal IS more PC1-concentrated. A real architectural difference.

k-sweep stable: ordering holds at every k (64 to 512), both architectures.

Whitening-dimension sweep: the fractal>control ordering is what must be stable, not the absolute rho

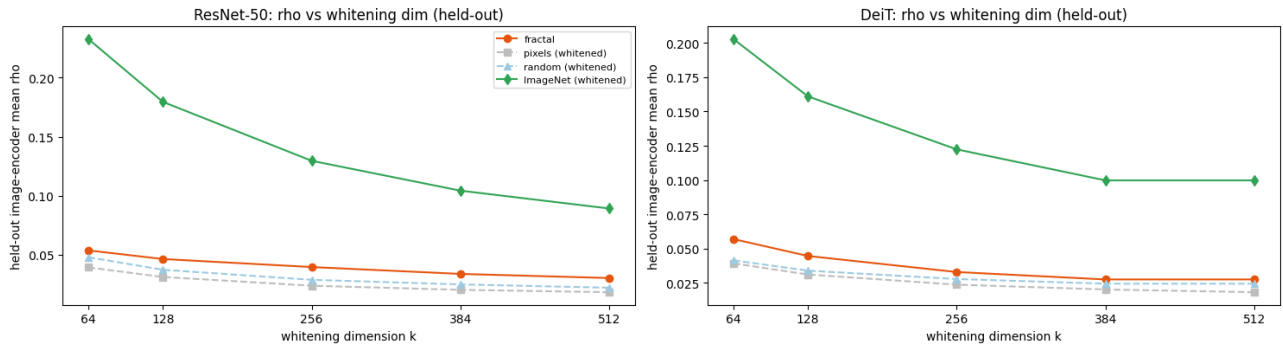


Figure G9.3 -- Whitening-dimension sweep. Absolute rho falls monotonically; fractal > control ordering holds at every k.

De-anisotropization heatmap: raw vs whitened (absolute rho falls; the ordering is the point)

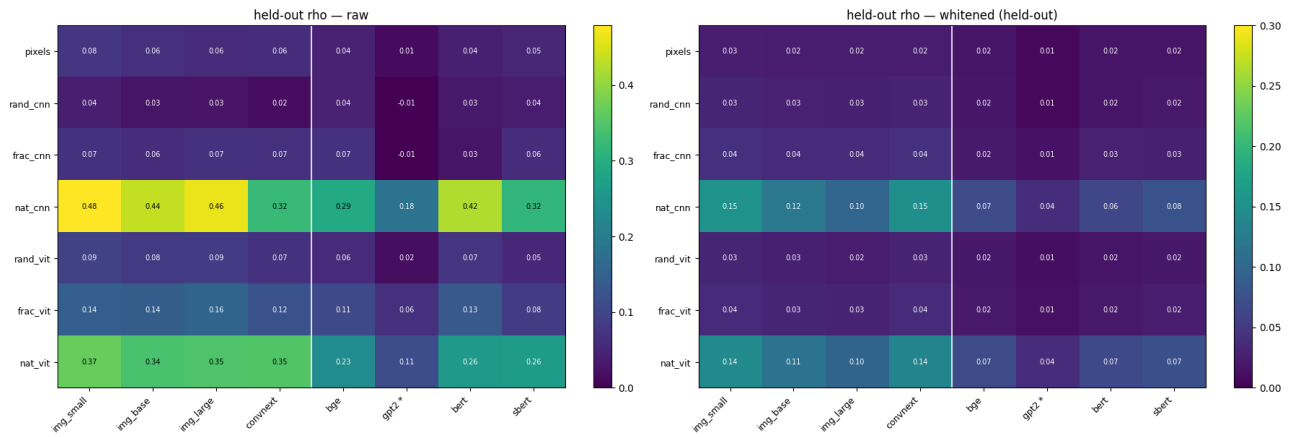


Figure G9.4 -- Raw vs whitened held-out rho. Different colour scales; the ordering, not the magnitude, is the point.

G10 -- Hub entry-map recoverability (the functional test)

G9 measured VISIBLE structure (raw shape agreement). G10 measures RECOVERABLE structure: a fitted linear entry map pushes fractal features into the frozen 512-d hub, and a frozen caption head (trained on DINOv2-small, never on fractals) reads them out. R@1 scored on 1,000 held-out rows against the bge retrieval gallery.

space	transfer R@1	native R@1	% native	at floor?
frac_cnn	0.014	0.014	n/a	YES
frac_vit	0.020	0.022	n/a	YES
rand_cnn	0.012	0.012	n/a	YES
rand_vit	0.017	0.018	n/a	YES
nat_cnn	0.258	0.274	94.2%	no
nat_vit	0.204	0.208	98.1%	no
pixels	0.000	0.004	n/a	YES

Effective transfer floor = 0.050 (max of 0.05 and 2x control 0.001). Fractal and random encoders are ALL at the floor; '% of native' is undefined there.

Pre-registered verdicts

hypothesis	CNN	ViT
H1 fractal > random & > pixels	FALSIFIED	FALSIFIED
H2 ImageNet > fractal	CONFIRMED	CONFIRMED
H3 recoverable >> visible	NOT MEANINGFUL	NOT MEANINGFUL

H1 FALSIFIED: fractal transfer (CNN 0.014, ViT 0.020) is indistinguishable from its random twin (0.012, 0.017) -- gaps of +0.002 and +0.004 at 0.005-0.006 SE. H3 NOT MEANINGFUL: when both fractal transfer and its native head are at the floor, their ratio (formerly '100%') is a ratio of two near-zero numbers and carries no information. The floor guard catches this.

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[vit] H1 fractal>random &gt;pixels: FALSIFIED (vs random d=+0.004/0.006SE, vs pixels
d=+0.021/0.004SE)

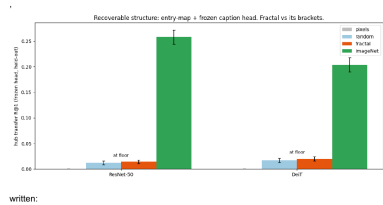
[vit] H2 ImageNet>fractal: CONFIRMED (d=+0.184/0.015SE); fractal recoverable = n/a (at
floor)

loaded G9 reality fractions R (visible structure)

[cnn] H3 NOT MEANINGFUL: fractal transfer 0.014 is at the floor (<= 0.050) or does not beat
random (H1 FALSIFIED). '% of native' is undefined; the fractal encoder's structure is NOT
recoverable into the hub.

[vit] H3 NOT MEANINGFUL: fractal transfer 0.020 is at the floor (<= 0.050) or does not beat
random (H1 FALSIFIED). '% of native' is undefined; the fractal encoder's structure is NOT
recoverable into the hub.

Reading: H3 only has meaning when fractal transfer CLEARS THE FLOOR and beats random
(H1 confirmed). At the floor, '100% of native' is 100% of ~nothing -- the correct conclusion is that
the fractal structure is NOT recoverable, matching H1. Natural-image encoders clear the floor
and are the meaningful recoverable-structure comparison.
```



written:

Figure G10.1 -- Hub transfer R@1 with bootstrap SEs. Fractal and random bars labeled 'at floor'; ImageNet bars at 0.20-0.26.

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G10_fractal_hub_transfer.png
G10_recoverable_vs_visible.png
G10_results.json
G10_transfer_heatmap.png
G10_transfer_table.csv
```

BOTTOM LINE: G9 asked if fractal structure is VISIBLE (raw geometry) -- mostly at the floor. G10 asks if it is RECOVERABLE (fitted entry map + frozen head). Answer: the fractal encoders transfer at the floor (~0.01-0.02 R@1), statistically indistinguishable from their random twins (H1 FALSIFIED). So the structure is NOT recoverable into the caption hub either -- a fitted map cannot rescue what is barely there. '% of native' reads ~100% only because the native head is ALSO at the floor; that ratio is meaningless and is guarded out. Natural-image encoders recover 0.20-0.26 and are the real point of comparison.

Figure G10.2 -- Transfer heatmap. '% native' column shows 'floor' for all at-floor rows. Only nat_cnn (94.2%) and nat_vit (98.1%) have meaningful percentages.

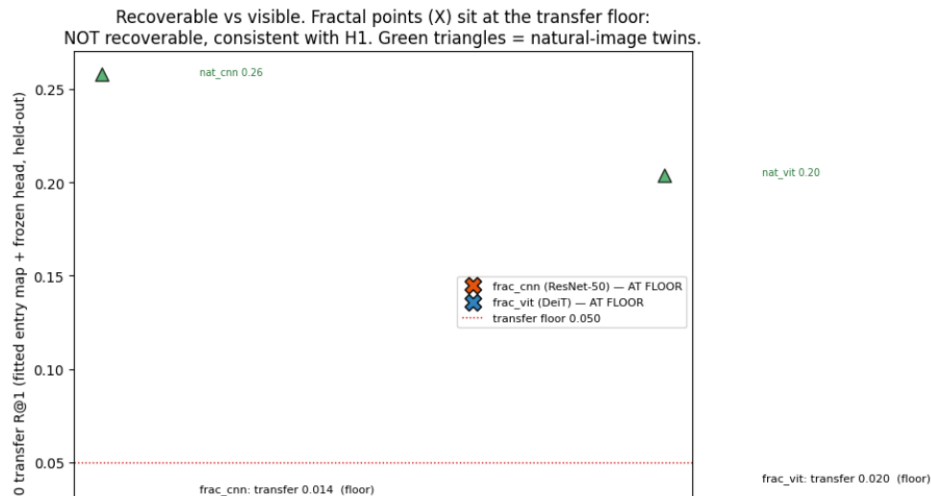


Figure G10.3 -- Recoverable (y, absolute transfer R@1) vs visible (x, G9 raw R). Fractal points (X markers) sit at the transfer floor; natural-image twins (green triangles) are the meaningful recoverable-structure comparison. The fractal structure is NOT recoverable.

Corrected bottom line

Across eight encoders, G8 finds statistically non-null, map-free relational agreement, strongest within model families and weaker across modalities. Much of the cross-modal agreement is carried by dominant anisotropic directions and falls after whitening (0.288 -> 0.062). G9 shows that FractalDB training produces a small shared geometric signal above random initialization -- directly visible for DeiT and visible for ResNet only after held-out de-anisotropization -- while natural-image training remains substantially stronger ($R \sim 10\text{-}20\%$ of the natural bracket). No reliable image-vs-language specialization is established (P3 INCONCLUSIVE). G10 shows the fractal structure is NOT functionally recoverable into the caption hub: fractal transfer sits at the floor, indistinguishable from random twins (H1 FALSIFIED). These results are consistent with training-data-driven convergence and a weak form of PRH, but do not establish a universal representation, image-versus-language specialization, or functional recoverability from fractal-only training.